Nonexistence of homogeneous rotation symmetric bent functions of degree four in twelve variables

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Abstract

We prove that no homogeneous rotation symmetric Boolean function of algebraic degree four in twelve variables is bent. The argument combines structural restrictions with a finite certificate verified in Lean. We first derive necessary conditions from restricted weight divisibility and from autocorrelations on periodic inputs. These conditions place every possible bent function in one of three explicit affine families. We then certify a refutation for each family using Boolean polynomial identities, affine substitutions, exhaustive branching, and Walsh witnesses at the terminal cases. A soundness theorem connects the concrete certificates to the original functions, yielding nonexistence for every choice of the forty-three coefficients in the quartic cyclic basis. The equivalence between the autocorrelation and Walsh formulations of bentness is also verified. The certificates, source files, and verification records are described in the accompanying data documentation. This establishes the twelve-variable quartic case of the homogeneous rotation symmetric bent function conjecture. Verification of the certificate does not require repeating the original exhaustive search.

Keywords: Boolean functions; bent functions; rotation symmetry; homogeneous functions; computer-assisted proof; Lean.

1 Introduction

Bent Boolean functions have Walsh coefficients of constant absolute value and consequently attain maximum distance from the affine Boolean functions [1]. Their existence is restricted to even numbers of variables. Rotation symmetry gives a natural class in which to investigate the interaction between this spectral property and algebraic structure: a single monomial determines all its cyclic translates, so a function can be specified by substantially fewer coefficients than an arbitrary algebraic normal form.

Stănică and Maitra [2] studied the enumeration and cryptographic properties of rotation symmetric Boolean functions and formulated the conjecture that homogeneous rotation symmetric functions of degree greater than two are not bent. Subsequent work established necessary conditions and nonexistence results for various parameters and support configurations; see, for example, Meng, Chen and Fu [3]. Sun, Shi, Liu and Fu [4] developed further restrictions using autocorrelations, distances between indices and derivatives. These approaches motivate the use of periodic inputs and algebraic constraints in the present work.

Rotation symmetry alone does not exclude bentness above degree two. Recent constructions of Polujan, Kudin and Pasalic [5] produce rotation symmetric bent functions outside the completed Maiorana–McFarland class. Their quartic family has $n = 2m$ with m odd and $3 \nmid m$, while their maximum-degree family has degree six when $n = 12$. Tang's preprint (Cryptology ePrint Archive 2026/1799, https://eprint.iacr.org/2026/1799) gives constructions in dimensions $30 \cdot 7^j$ and $70t$, neither of which includes twelve. These existence results concern a different question from homogeneous quartic nonexistence. Here homogeneity refers to the whole algebraic normal form, rather than just its maximum degree.

For twelve variables, the prime-factor hypotheses in Theorems 1–2 of [4] are not satisfied. Their Theorems 3 and 4(1) retain restrictions on monomial supports, the degree threshold in Theorem 4(2) is five, and Theorems 5–6 concern cubic functions. Our result concerns the full quartic cyclic space at this dimension.

We consider the complete space generated by the cyclic orbits of quartic monomials in twelve variables. There are forty-three such orbits. Thus the question concerns all 2^{43} coefficient choices, with no initial restriction to a selected collection of monomials.

Theorem 1.1. No homogeneous rotation symmetric Boolean function of degree four in twelve variables is bent.

The proof is assisted by computation. Its structural part reduces bentness to three affine families of dimension thirty-six, together with explicitly defined quadratic equations. Its finite part supplies refutations of these three systems relative to the bentness predicate. The accompanying independent certificate checker also provides a practical route to reproducing the finite exclusions without rebuilding the Lean dependency graph. The refutations use polynomial identities and affine substitutions; terminal observations are rejected by exact Walsh witnesses. A general soundness argument explains why their acceptance excludes every possible bent function in the original space. This argument and the three concrete refutations have been checked in Lean 4.35.0.

The contribution is a certified finite case of the conjecture, together with an explicit reduction that separates necessary structural conditions from the final exclusion. In dimension twelve, the quartic cyclic space already has 2^{43} elements. A certificate makes the logical coverage of that space inspectable without treating a reported search count as the proof.

The distinction between a search and a certificate is useful here. An exhaustive search can suggest nonexistence, but its final count alone does not explain coverage or the validity of its rejection tests. The certificate used below makes both explicit. Its verification does not rely on trusting the program that discovered the decomposition, or on repeating an enumeration of all 2^{43} functions.

Related work by the author concerns antipodal necessity in degree at most three and the homogeneous cubic case (preprint, https://doi.org/10.6084/m9.figshare.34165260.v2). The present quartic argument is self-contained and does not use that result as a hypothesis.

Section 2 defines the family and the spectral conventions. Section 3 derives the necessary conditions and the three affine families. Section 4 describes the refutation rules and proves their soundness. Section 5 connects the accepted concrete certificates to Theorem 1.1. Section 6 discusses the scope and supplementary records.

2 Preliminaries and the quartic cyclic space

All polynomial arithmetic is over $\mathbb{F}_2$ unless otherwise stated. Let $V = \mathbb{F}_2^{12}$, with coordinates numbered from 0 to 11. Write ρ for cyclic rotation by one position. A Boolean function $f : V \to \mathbb{F}_2$ is rotation symmetric if $f \circ \rho = f$. Its algebraic normal form is the unique square-free polynomial representing it. It is homogeneous of degree four if every monomial in this form has degree four.

Let $\mathcal{O}_0, \ldots, \mathcal{O}_{42}$ be the ordered cyclic orbits of four-element subsets listed in Appendix A. Define

$$p_j(x) = \sum_{S \in \mathcal{O}_j} \prod_{i \in S} x_i, \qquad f_c(x) = \sum_{j=0}^{42} c_j p_j(x), \quad c \in \mathbb{F}_2^{43}. \tag{2.1}$$

Proposition 2.1. The family (2.1) consists of the zero function and all homogeneous rotation symmetric functions of degree four on V, each represented exactly once.

Proof. Rotation acts on monomial supports. Uniqueness of algebraic normal form implies that rotation invariance is equivalent to constancy of the coefficient on each support orbit. The forty-three orbits in Appendix A are pairwise disjoint and cover the $\binom{12}{4} = 495$ four-element supports. Assigning one coefficient to each orbit therefore gives every invariant quartic form. Disjointness makes the representation unique. Every nonzero combination has a degree-four monomial and hence has degree exactly four. □

For $a, u \in V$, define the Walsh transform and autocorrelation by

$$W_f(a) = \sum_{x \in V} (-1)^{f(x)+a \cdot x}, \qquad C_f(u) = \sum_{x \in V} (-1)^{f(x)+f(x+u)}. \tag{2.2}$$

The function is bent when $|W_f(a)| = 64$ for every a.

Lemma 2.2. A Boolean function on V is bent if and only if $C_f(u) = 0$ for every nonzero $u \in V$.

Proof. Expanding the square of $W_f(a)$ and putting $y = x + u$ gives

$$W_f(a)^2 = \sum_{u \in V} C_f(u)(-1)^{a \cdot u}. \tag{2.3}$$

Since $C_f(0) = 4096$, vanishing at nonzero directions implies $W_f(a)^2 = 4096$. Conversely, Fourier inversion applied to (2.3) gives

$$C_f(u) = 2^{-12} \sum_{a \in V} W_f(a)^2 (-1)^{a \cdot u}.$$

If the squares are all 4096, character orthogonality makes this zero for nonzero u. □

3 Necessary conditions and affine reduction

3.1 Quadratic conditions from restricted weights

For an eight-element coordinate set S, let V_S be the coordinate subspace supported on S. Orthogonality gives the restriction identity

$$\sum_{x \in V_S} (-1)^{f(x)} = \frac{1}{16} \sum_{a \in V_S^\perp} W_f(a). \tag{3.1}$$

Lemma 3.1. If f is bent, then the weight of $f|_{V_S}$ is divisible by four.

Proof. There are sixteen frequencies in $V_S^\perp$. For a bent function the right side of (3.1) is four times a sum of sixteen signs, and is therefore divisible by eight. The left side equals $256 - 2\,\mathrm{wt}(f|_{V_S})$. The stated divisibility follows. □

For bits t_x, the second binary digit of the integer $\sum_x t_x$ is $\sum_{x<y} t_x t_y$ in $\mathbb{F}_2$: both are $\binom{\sum_x t_x}{2}$ modulo two. Applying this to $t_x = f_c(x)$ produces a polynomial $Q_S(c)$ of degree at most two, after reducing $c_i^2 = c_i$. Lemma 3.1 implies $Q_S(c) = 0$ whenever f_c is bent.

This definition is explicit and does not require interpolation. If $v_i = (p_i(x))_{x \in V_S}$, its coefficients are

$$[c_i]Q_S = \lfloor |v_i|/2 \rfloor \bmod 2, \qquad [c_i c_j]Q_S = (|v_i| \bmod 2)(|v_j| \bmod 2) + |v_i \cap v_j| \bmod 2 \quad (i < j). \tag{3.2}$$

For the second formula, the coefficient is the sum of $v_i(x)v_j(y)$ over all ordered pairs with $x \neq y$, which equals the displayed expression. The first formula counts pairs of ones in v_i.

Only a necessary consequence of weight divisibility is retained in each Q_S: vanishing of its second binary digit. We do not assert that a common zero of these equations is bent.

There are forty-three cyclic orbits of eight-element subsets, obtained by complementing the quartic support orbits. Rotation invariance makes one equation per orbit sufficient. Write $Q_0, \ldots, Q_{42}$ for the resulting equations, ordered by the eight-coordinate support masks in Appendix B. The filter numbering is independent of the quartic orbit numbering.

3.2 Linear consequences

Define

$$\begin{aligned} R_1 &= c_1 + c_7 + c_9 + c_{18} + c_{24} + c_{32} + c_{34} + c_{36}, \\ R_2 &= c_{26} + c_{40} + c_{42}, \\ R_3 &= c_0 + c_1 + c_2 + c_6 + c_7 + c_9 + c_{12} + c_{15} + c_{18} + c_{23} + c_{24} + c_{25} + c_{32} + c_{34} + c_{36} + c_{37}. \end{aligned} \tag{3.3}$$

Lemma 3.2. Every common zero of the forty-three quadratic conditions satisfies $R_1 = R_2 = R_3 = 0$.

Proof by explicit polynomial identities. In the Boolean polynomial ring

$$\mathbb{F}_2[c_0, \ldots, c_{42}]/(c_i^2 + c_i),$$

each R_i is a sum of products of a filter Q_j with either 1 or a coefficient variable. The three lists have respectively twenty-two, one and twenty-four products. Their full factors are displayed in Appendix B and supplied in `quartic_audit_2026_10_17/linear_law_certificates.json`; the filters are reconstructed from (3.2). Multiplication and cancellation of equal monomials give (3.3), as checked in the formal filter linkage. Evaluation at a common zero makes each product zero. □

Thus (3.3) is derived from necessary conditions; it is not an additional assumption restricting the family under investigation. The explicit identities form part of the finite certificate.

3.3 Restriction to period four

Let $r(y) = (y, y, y)$ for $y \in \mathbb{F}_2^4$. Direct restriction of the orbit sums gives

$$f_c(r(y)) = \alpha A(y) + \beta B(y) + \gamma C(y) + \delta D(y), \tag{3.4}$$

where

$$\begin{aligned} A &= y_0y_1 + y_1y_2 + y_2y_3 + y_3y_0, \\ B &= y_0y_2 + y_1y_3, \\ C &= y_0y_1y_2 + y_0y_1y_3 + y_0y_2y_3 + y_1y_2y_3, \\ D &= y_0y_1y_2y_3. \end{aligned}$$

Here $\alpha = c_{15} + c_{18} + c_{19} + c_{23} + c_{24}$, $\beta = c_{40}$, $\delta = c_{42}$, and

$$\begin{aligned} \gamma &= \sum_{j \in J} c_j, \\ J &= \{1,2,3,5,6,7,8,9,11,12,13,16,17,20,21,22, \\ &\qquad 25,27,28,29,30,32,34,36,38,39,41\}. \end{aligned} \tag{3.5}$$

Lemma 3.3. If f_c is bent, then the tuple $(\alpha, \beta, \gamma, \delta)$ is one of

$$(0,1,0,0), \quad (1,1,0,0), \quad (0,0,0,1), \quad (1,1,1,1). \tag{3.6}$$

Proof. Put $g = f_c \circ r$. Rotation by four coordinates has order three. For each nonzero $u \in \mathbb{F}_2^4$, the summand defining $C_{f_c}(r(u))$ is constant on its orbits. There are sixteen fixed points, precisely the $r(y)$, and all other orbits have size three. Therefore

$$C_{f_c}(r(u)) \equiv C_g(u) \pmod 3. \tag{3.7}$$

Lemma 2.2 implies that every $C_g(u)$ is divisible by three. Testing (3.4) for its sixteen coefficient tuples gives the following complete list. Truth tables use the integer whose bit y is $g(y)$, with y encoded in increasing binary order.

$\alpha + 2\beta + 4\gamma + 8\delta$	Tuple	Truth table
2	$(0,1,0,0)$	`0x6ca0`
3	$(1,1,0,0)$	`0x7ee8`
8	$(0,0,0,1)$	`0x8000`
15	$(1,1,1,1)$	`0x9668`

This table is obtained by summing the sixteen signs for each of the fifteen nonzero directions for each tuple. The finite classification, restriction (3.4), and orbit descent (3.7) are checked in the formal development. □

Lemma 3.4. A bent f_c cannot have the last tuple in (3.6).

Proof. Let Q be the filter for the eight-coordinate support with binary mask `0x3cf`. Formula (3.2) gives

$$Q = L + c_{26} + c_{40} + c_{26}c_{42}, \qquad L = c_0 + c_2 + c_6 + c_{12} + c_{15} + c_{23} + c_{25} + c_{37} = R_1 + R_3.$$

The Boolean identity

$$c_{40}c_{42} = Q + R_1 + R_3 + (1 + c_{42})R_2 \tag{3.8}$$

follows by expansion. All terms on the right vanish for a bent function. Since $\beta = c_{40}$ and $\delta = c_{42}$, the tuple $(1,1,1,1)$ contradicts (3.8). □

3.4 Three affine families

For each of the first three tuples in (3.6), impose its four values and the three equations (3.3). These seven independent affine equations define a family $\phi_k(\mathbb{F}_2^{36})$, where $k = 0, 1, 2$ corresponds respectively to codes $2, 3, 8$.

An explicit parametrization avoids a rank calculation hidden in software. Choose freely the coordinates

$$I = \{0, 1, \ldots, 23, 25, 27, 28, 29, 30, 31, 32, 33, 34, 35, 38, 39\}.$$

Determine the other coordinates in the following order:

$$\begin{aligned}
c_{24} &= \alpha + c_{15} + c_{18} + c_{19} + c_{23},\\
c_{40} &= \beta, \qquad c_{42} = \delta, \qquad c_{26} = \beta + \delta,\\
c_{36} &= c_1 + c_7 + c_9 + c_{18} + c_{24} + c_{32} + c_{34},\\
c_{37} &= c_0 + c_1 + c_2 + c_6 + c_7 + c_9 + c_{12} + c_{15} + c_{18} + c_{23} + c_{24} + c_{25} + c_{32} + c_{34} + c_{36},\\
c_{41} &= \gamma + \sum_{j \in J \setminus \{41\}} c_j.
\end{aligned} \tag{3.9}$$

Each right side uses only free or previously determined coordinates. Projection onto I is the inverse of this parametrization on the family. In particular, each family has dimension thirty-six.

Proposition 3.5. If f_c is bent, then there exist $k \in \{0, 1, 2\}$ and $u \in \mathbb{F}_2^{36}$ such that

$$c = \phi_k(u), \qquad Q_j(\phi_k(u)) = 0 \quad (0 \le j < 43). \tag{3.10}$$

Proof. Lemmas 3.1–3.4 give the seven equations for one of the three tuples. Formula (3.9) reconstructs c from its free coordinates. The filter conditions follow from Lemma 3.1. □

4 Refutation certificates

Write $q_{k,j} = Q_j \circ \phi_k$. Exact substitution gives forty-three Boolean quadratic polynomials in thirty-six variables for each family. The coefficient observation is the affine map ϕ_k. A state records a further affine substitution E on these variables. Its interpretation consists of the equations $q_{k,j}(E(z)) = 0$ and the observed coefficient vector $\phi_k(E(z))$.

The assertion proved by a refutation at this state is

$$\forall z \in \mathbb{F}_2^{36}, \quad \big(q_{k,j}(E(z)) = 0 \text{ for every } j\big) \implies \neg\,\mathrm{Bent}(f_{\phi_k(E(z))}).$$

The certificate records the data needed to establish this assertion recursively, using the following rules. All polynomial equalities are equalities in the Boolean polynomial ring, so repeated variable factors are reduced using $z_i^2 = z_i$.

Contradiction. A specified sum of the current equations is the constant polynomial one. No common zero exists.

Elimination. A specified sum of the equations is $z_i + a(z)$, where a is affine. Any common zero satisfies $z_i = a(z)$. Substitute this equality into the recorded environment and continue with the child refutation.

Branching. For an affine a, use the two substitutions $z_i = a(z)$ and $z_i = a(z) + 1$. Every assignment satisfies one of these two equalities. Both resulting states must have refutations.

Terminal rejection. The composed coefficient observation depends only on an explicitly checked finite set of active coordinates. Enumerate their assignments. For each resulting coefficient vector c, the certificate supplies a frequency $b \in V$ such that

$$W_{f_c}(b)^2 \neq 4096. \tag{4.1}$$

The Walsh values are evaluated with exact integer arithmetic; no floating-point threshold or probabilistic rejection is used. Every observation at that terminal is therefore non-bent. The support check on the observation is essential: it justifies that varying the enumerated

active coordinates covers every observation of the state, including those from assignments not satisfying the current quadratic equations.

Proposition 4.1 (soundness). An accepted refutation at a state excludes bentness of every observed function whose parameters satisfy the equations at that state.

Proof. Induct on the refutation. At a contradiction node, evaluation at a common zero would give $0 = 1$. At an elimination node, the certified polynomial identity forces the substitution; the updated environment has the same value at the assignment, so the child applies. At a branching node, one of the two affine equalities holds, and the corresponding child applies. At a terminal, the checked support condition identifies the observation with one obtained from an enumerated active assignment. Its verified witness (4.1) contradicts bentness. □

For example, if the sum of two current equations is $z_3 + z_0 + 1$, any common zero has $z_3 = z_0 + 1$ and can be passed to the elimination child. This is a deduction from equations, rather than deletion of the branch $z_3 = z_0$. Branching, when used, retains both possible values. Keeping these operations distinct is necessary for coverage.

The implementation stores polynomials and affine maps as packed bit representations. Its evaluation and substitution lemmas connect those representations to the interpretation above. Terminal lookup tables are accompanied by proofs that their stored coefficient keys have valid Walsh witnesses. Consequently a successful lookup does not introduce an assumption that the listed functions are non-bent.

5 Certified verification and the main theorem

5.1 Concrete acceptance

Proposition 5.1 (finite certified verification). For each $k = 0, 1, 2$, the supplied certificate is an accepted refutation of the equations $q_{k,j} = 0$ with coefficient observation ϕ_k.

This proposition is the computational part of the proof. The concrete instances are `H12CloudChart0.refutation`, `H12CloudChart1.refutation`, and `H12CloudChart2.refutation`. Each is a proof object of the refutation predicate described in Section 4. They are combined without an additional hypothesis in `H12Final.all_chart_refutations`.

The concrete instances were checked in Lean 4.35.0 and composed with the general soundness and coverage theorems. The final axiom reports contain only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. The final theorem therefore does not assume the correctness of a certificate generator or a search result. Lean's proof architecture is described in [6]. Appendix C identifies the preserved sources and verification records.

5.2 Completion of the proof

Proof of Theorem 1.1. By Proposition 2.1, write a putative bent function as f_c. Proposition 3.5 places c in one of the three affine families and gives a common zero of its necessary quadratic equations. Proposition 5.1 supplies the corresponding refutation. Proposition 4.1 excludes bentness of its observation, which is exactly f_c. This is a contradiction. □

The exact final formal statement is

```
theorem H12Final.no_bent_quartic (c : BitVec 43) :
    ¬ H12Descent.BentAC (H12.quarticFunction c)
```

Here `BitVec43` represents $\mathbb{F}_2^{43}$, and the concrete function evaluates the orbit sums (2.1) on twelve-bit inputs. `BentAC` is the autocorrelation condition of Lemma 2.2. The formal theorem `H12Fourier.bent_iff_flat` proves its equivalence with Walsh flatness on this domain. Thus the result also gives

$$\forall c \in \mathbb{F}_2^{43}\ \exists b \in V : \quad W_{f_c}(b)^2 \neq 4096. \tag{5.1}$$

The concrete support orbits used by the formal theorem are the orbits in Appendix A. Their disjointness and coverage give the mathematical identification in Proposition 2.1. The implementation and source audit for this identification are described in Appendix C.

5.3 Use of AI-assisted tools

During this research and the preparation of the manuscript, generative AI tools (OpenAI Codex and Gemini) were used to support mathematical discussion and manuscript preparation. Codex also assisted with programming and Lean formalization. The formal nonexistence statement was verified by the Lean kernel using the certificates described in this paper. Responsibility for the scientific content and the final manuscript rests with the author.

6 Conclusion and availability of verification material

Theorem 1.1 settles the quartic case in twelve variables by combining structural reductions with certified finite refutations. The necessary weight conditions yield a quadratic system; periodic autocorrelations and its linear consequences reduce coverage to three explicit affine families. The certificate then excludes every bent observation in those families.

The result concerns degree four in twelve variables. It does not establish nonexistence in sixteen variables, all even dimensions, or the whole space of functions of degree at most four. Extending the method requires new reductions or new certified refutations for the parameters concerned. A low-degree restriction to a subspace alone is not an obstruction to bentness; the proof depends on the specific necessary equations and their refutations.

The accompanying material provides two routes to verification: an independent replay of the finite certificate using standard Python, and the complete mathematical source archive for the formal proof. The former avoids a reconstruction of the Lean dependency graph; the latter preserves the exact sources of the kernel-checked result. Appendix C states their scope and reproduction requirements.

Appendix A: Ordered quartic support orbits

Each row gives the representative support and the orbit size for the coefficient index used in (2.1). Indices are zero based and all translates are modulo twelve. The orbit sizes total $40 \cdot 12 + 2 \cdot 6 + 3 = 495$.

Index	Representative support	Orbit size
0	$\{0, 1, 2, 3\}$	12
1	$\{0, 1, 2, 4\}$	12
2	$\{0, 1, 2, 5\}$	12
3	$\{0, 1, 2, 6\}$	12
4	$\{0, 1, 2, 7\}$	12
5	$\{0, 4, 5, 6\}$	12
6	$\{0, 3, 4, 5\}$	12
7	$\{0, 2, 3, 4\}$	12
8	$\{0, 1, 3, 4\}$	12
9	$\{0, 1, 3, 5\}$	12
10	$\{0, 1, 3, 6\}$	12
11	$\{0, 1, 3, 7\}$	12

Index	Representative support	Orbit size
12	$\{0,4,5,7\}$	12
13	$\{0,3,4,6\}$	12
14	$\{0,2,3,5\}$	12
15	$\{0,1,4,5\}$	12
16	$\{0,1,4,6\}$	12
17	$\{0,1,4,7\}$	12
18	$\{0,1,4,8\}$	12
19	$\{0,3,4,7\}$	12
20	$\{0,2,3,6\}$	12
21	$\{0,1,5,6\}$	12
22	$\{0,1,5,7\}$	12
23	$\{0,1,5,8\}$	12
24	$\{0,3,4,8\}$	12
25	$\{0,2,3,7\}$	12
26	$\{0,1,6,7\}$	6
27	$\{0,2,6,7\}$	12
28	$\{0,3,6,7\}$	12
29	$\{0,4,6,7\}$	12
30	$\{0,2,5,6\}$	12
31	$\{0,3,5,6\}$	12
32	$\{0,2,4,5\}$	12
33	$\{0,2,4,6\}$	12
34	$\{0,2,4,7\}$	12
35	$\{0,2,4,8\}$	12
36	$\{0,3,5,7\}$	12
37	$\{0,2,5,7\}$	12
38	$\{0,2,5,8\}$	12
39	$\{0,3,5,8\}$	12
40	$\{0,2,6,8\}$	6
41	$\{0,3,6,8\}$	12
42	$\{0,3,6,9\}$	3

Appendix B: Filter ordering and linear certificates

A mask s denotes the coordinate set $\{i : \mathrm{bit}_i(s) = 1\}$. In the following order, Q_j is obtained from that support using (3.2):

$$255, 383, 639, 509, 447, 703, 765, 831, 1013, 507, 479$$
$$735, 893, 863, 1375, 763, 927, 1003, 1005, 503, 495, 751$$
$$957, 879, 1391, 891, 943, 1455, 1403, 759, 975, 983, 987$$
$$989, 887, 1399, 955, 951, 1463, 1467, 1495, 1499, 1755.$$

Define $m_{-1} = 1$ and $m_v = c_v$ for $0 \le v < 43$. Each row below encodes the exact identity $R_i = \sum_{(j,v)\in T_i} m_v Q_j$. The products are in the Boolean polynomial ring.

For R_1, the pairs in T_1 are

$$(1,7), (3,1), (5,32), (8,9), (10,5), (12,13), (13,30)$$
$$(15,20), (17,16), (19,3), (21,29), (24,36), (26,27), (28,34)$$
$$(31,22), (33,11), (34,24), (36,18), (38,41), (40,-1), (40,40)$$
$$(41,38).$$

For R_2, the pairs in T_2 are

$$(42, -1).$$

For R_3, the pairs in T_3 are

$$(30, -1), (30, 26), (30, 40), (30, 42), (40, -1), (40, 26), (40, 40)$$
$$(40, 42), (42, 0), (42, 1), (42, 2), (42, 6), (42, 7), (42, 9)$$
$$(42, 12), (42, 15), (42, 18), (42, 23), (42, 24), (42, 25), (42, 32)$$
$$(42, 34), (42, 36), (42, 37).$$

In particular, $R_2 = Q_{42}$. The filter used in (3.8) is Q_{30}, since its support mask is $975 = \texttt{0x3cf}$. These conventions make every linear identity and the excluded periodic case reproducible directly from the orbit table and (3.2).

Appendix C: Verification material and formal scope

The preserved Lean result is the universally quantified statement in Section 5.2, together with `no_flat_quartic` and `walsh_witness`. The final integration used Lean 4.35.0 at repository commit `344ab2e1b330d4a6be3f00603e9423db39cf3354` and completed on 21 October 2026. The final modules were checked with `leanchecker`. The reports for the combined refutations and three final consequences contain only `propext`, `Classical.choice`, and `Quot.sound`; neither `sorryAx` nor `ofReduceBool` occurs in these dependency reports.

The source explicitly defines the 43 support orbits and the coefficient-to-function map. Their independent source audit checks all 495 quartic supports. The final module does not add a separate abstract representation theorem for arbitrary algebraic normal forms: Proposition 2.1 states the mathematical identification with this concrete basis.

Material	Purpose	Verification boundary
Article and appendices	Structural reductions and soundness argument	Includes all orbit representatives and linear-identity factors
Independent Python package	Checks affine decomposition and terminal Walsh witnesses	Separate four-root certificate; not a Lean proof
Complete Lean source archive	Preserves the formal argument and concrete root proofs	All 2093 mathematical sources match certified hashes
Final reports and logs	Identify the checked statements and logical dependencies	Records of the completed kernel integration

The independent checker reconstructs the basis, necessary linear and periodic conditions, and polynomial substitutions, then checks the finite decomposition and the exact Walsh witnesses. Its four initial affine families include the analytically contradictory fourth case. Its node and solution counts are not counts for the three Lean root packages. On the copy prepared for this preprint, replay completed in approximately 139 seconds; this is an observed time on one machine, not a hardware-independent guarantee. Its assertions are part of the checker, so it must be run without Python's optimization flag.

The formal source archive contains the 2091 previously certified distinct dependency modules and the two final modules, `H12Fourier` and `H12Final`. Source hashes are compared with the certified records and import closure is checked. This module count describes package size; it is not a percentage of mathematical coverage. The archive includes minimal Lake packaging.

A full fresh rebuild of that packaging has not been performed during preprint preparation. Such a rebuild can take hours and is distinct from replaying the compact independent certificate.

The verification repository provides the preserved sources, audits and dependency pins. A fixed record of the final evidence identifies the completed integration. The source archive and compact verifier accompany this preprint as separate files, so their availability does not depend on temporary execution caches. The full final binary artifact is identified by SHA256 `2cc9b4aadfe6bfd5f89721b5a72f9768854f4d46c475763dd29953f8153b331c`.

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